BIO 3105
Proposed Question Bank
Summer 2022

Introduction

1. Define biological engineering. Slide 3
2. Clearly state the value of biology knowledge for computer science students. Slide 11
3. Ecosystems and evolution are intertwined. Use an example to clarify. Slide 6, 7, 8
4. Identified the main distinctions between an engineer and a scientist. Slide 13
5. Mention the fields in modern biology where you can apply machine learning. Slide 34-40
6. Name three environment friendly ideas that can be implemented as business models. Slide 10
7. How a degraded ecosystem affects the ecosystem core? Give logical explanation. Slide 9
8. Suppose you have a pond where you want to grow fishes. Can you design a biorobot that can potentially help you to reduce the expenditure on manpower? Is that possible to use your expertise in this case? Slide 23, 24
9. Do you think you would be able to give your input as a computer engineer in tissue engineering? Design a probable project in that area with a few words. Slide 35
10. How you would differentiate "organ-on-a-chip" and "tissue engineering" as both of these techniques are used to give you new organs? Slide 39
11. Can you design a project in the field of Brain and Neuroscience using your own background? Explain briefly about the project and how you can implement your expertise there. Slide 34, 37

Cell Structure

1. Clarify why eukaryotic cells have flexible cell membranes. Slide 16
2. Recognize the differences between prokaryotic and eukaryotic cells. Slide12
3. Which organelle in the cell produces energy? Explain it. Slide 40
4. Describe how nutrients are transported across cell membranes. Slide 27-33
5. Compare the vacuoles of animals and plants. Slide 42
6. Which organelle do you think is most important for the living world? Write down the functions of that certain organelle. Slide 40, 41
7. Which organelle works in biogenesis? Mention its functions. Slide 37
8. Do you think cell membrane works in transportation? Briefly explain the procedure involved in it. Slide 27-33
9. Do you think mitochondrion allows transportation through it? If so, discuss the process briefly. Slide 27-33
10. Mention the functions of the substance you think prevents the cell from shrinking. Slide 49
11. Sketch the differences, as well as similarities between Lysosomes and Vacuoles. Slide 43, 45
12. Name the functions of the organelle you think necessary for cell division. Slide 48
13. Which organelle works in the aging process? Mention its characteristics. Slide 45, 46
14. Do you think transportation is possible through Lysosomes' membrane? Give details. Slide 27-33
15. Analyse why Bacteria and archaea are the prokaryotic?
16. State the importance of ergastics substance in the cell.

17. Which organelles are named suicidal bag/ power house? Clarify.
18. Importance of the cell membrane structure.

DNA and RNA

1. Mention the differences, as well as similarities between DNA and RNA. Slide 2,3
2. Sketch the constituents of nitrogenous bases used in both DNA and RNA, and show their differences in that pictorial view. Slide 6
3. Mention the characteristics of genetic code. Slide 14
4. Name the possible combinations of genetic code where only one pyrimidine is fixed in the first position of triplets. Slide 14
5. Do you think RNA could be your genetic material? Give logic behind your answer. Slide 12, 13
6. What codons can you have after fixing Adenine in the first place and Cytosine and Uracil in the second place? Slide 14
7. How do you think a DNA of some meters can be fitted inside a cell of nm scale? Slide 9
8. Sketch a diagram for the central dogma of life. Slide 9
9. Is there any significance of a nucleotide? Give proper reasons. Slide 8
10. How many base pairs of a 2.2m long DNA has? Slide 8
11. Elaborate the processes Transcription and Translation, and describe the significance mRNA tRNA in these processes.

Cell Division

1. Show pictorial view of the differences, as well as similarities between Metaphase and Metaphase I. Comment on the genetic variations that can be introduced via Metaphase I. Slide 27
2. Show pictorial view of the phases where you might have the probabilities of tumor, or in some cases cancer. [Hint: Cell cycle for a day would be good to show this.] Slide 14-16
3. Do you think some genetic disorder can be carried through all the gametes from Meiosis? Give details in a sample diagram. Slide 46
4. Mention the relations between aneuploidy and down syndrome. Slide 47
5. Mention the relations between aneuploidy and polyploidy. Slide 47-49
6. (a) Suppose you have Persian cat with a variation of dark brown eye, which is a dominant trait, and another one of light brown eyes. In the second generation what would be the percentage of dark brown eyes? (b) If you had the genotype ratio of pure dark brown, mixed dark brown, and pure light brown to be 1:2:1, can you trace the parents' traits? Slide 55
7. Mention the differences, as well as similarities between phenotype and genotype with proper examples. Slide 53-55
8. Structural unit of living organisms. What are the three basic units need to form of cell. [Hint: nucleus, cytoplasm, membrane]
9. Clarify the importance of check points. Slide 14-16

10. Mention where nuclear envelop dissolves and form in the mitosis. Slide 12
11. Differentiate the anaphase of meiosis 1 and mitosis. Slide 36
12. Why there is no interphase in meiosis 2? In which phase most of the cells remain? Slide 32
13. What happened in the S phase in meiosis 2? Slide 32
14. Explain Interphase.
15. Chromosomal abnormalities (structural and numerical).